

Paper 8.75 - Lattice Closure as Next-State Precondition

CQE-paper-08.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-08.75.md

Paper 8.75 - Lattice Closure as Next-State Precondition

Purpose

Paper 8.75 explains how the lattice closure template becomes the block wrap for Papers 1-8 and the precondition for Paper 9.

Exported State

Paper 8 exports:

```
local chain (1,3,7,8,24)
powered terminal 72 = 8 x 9
sheet K bound = 9
Fano/Hamming verified rung
extended Hamming/E8 seed verified rung
Golay ingredient verified rung
Gamma72 transport boundary
Leech/Gamma72 landing audit boundaries
```

Wrap Back to Paper 1

Paper 8 closes the first local block by showing that the Paper 1 carrier can enter a higher-dimensional closure scaffold without changing its proof status.

The wrap is:

```
8-state LCR chart -> dimension-8 seed -> 24 = 3 x 8 -> 72 = 8 x 9
```

The wrap does not retroactively prove unclosed landing claims.

Use in Paper 9

Paper 9 may treat Paper 8 closure outputs as frame inputs for Hamiltonian window emergence. The admissible input is the verified scaffold and sheet bound, not the unproved Leech or Gamma72 landing.

The allowed transition is:

```
verified closure scaffold -> Hamiltonian window frame
```

The disallowed transition is:

```
unproved global landing -> closed time/dynamics theorem
```

Use in Later Papers

Later papers may use Paper 8 for:

- lattice-code closure frames,
- powered sheet bounds,
- Golay ingredient routing,
- Gamma72 transport boundaries,
- product-scale closure templates.

Every use must state whether it imports a verified local rung or a named audit boundary.

Precondition Rule

Before a later paper uses Paper 8, it should be able to answer:

```
which rung is imported?  
what verifier proves it?  
Is the claim local closure or global landing?  
what audit boundary remains open?  
which receipt is attached?
```

Conclusion

Paper 8.75 turns the first-block lattice scaffold into portable state. It closes the local block and gives Paper 9 a verified frame without smuggling in unproved global lattice landings.